DEEP LEARNING PROGRAMMING FINAL PROJECT

D3Net-based All-in-One Restoration for Smartphone Images with Dirty Lenses

SIDL adaptation, paper-baseline comparison, patch-size ablation, and failure analysis

20223081 신준철

Deep Learning Programming

MAIN RESULT
24.55 dB / 0.8507
Official leaderboard test result: 256 crop, epoch 73

COMPARISON SCOPE
AirNet / DiffUIR / D3Net
SIDL paper test results vs. our official submission

BEST SETTING
256×256 crop
Submitted checkpoint: epoch 73, 771 GMACs, 43.8M params

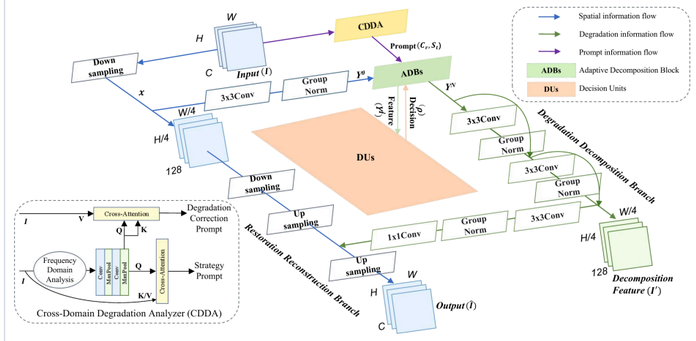
MAIN BOTTLENECK
Hard Dust / Mixed
Hard Fingerprint, Mixed, and Water remain difficult

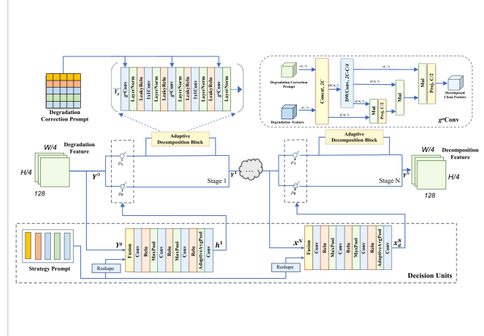
1. Problem & Dataset

SIDL is a real-world paired dataset for restoring smartphone images captured through contaminated lenses. The task is different from ordinary denoising or deblurring because lens contamination produces both local artifacts and global contrast degradation.

Data	512×512 patchified degraded-clean pairs
Validation	Type × difficulty split, PSNR / SSIM
Goal	Analyze D3Net adaptation, not claim SIDL SOTA

2. Method: D3Net Adaptation





D3Net predicts a residual correction: restored image = input + residual. Frequency-domain cues and dynamic decomposition blocks allow degradation-aware processing for mixed and local contamination patterns.

3. Experimental Setup

Model	D3Net, scratch training, 12 ADB blocks
Main run	256×256 random crop, batch 8, LR 1e-4
Submit	Epoch 73, 771 GMACs, 43.8M params
Loss / Opt.	L1 loss, Adam β=(0.9, 0.999), cosine LR
Validation	512×512 center crop, image-wise PSNR / SSIM
Scope	Limited-budget adaptation, not SIDL SOTA claim

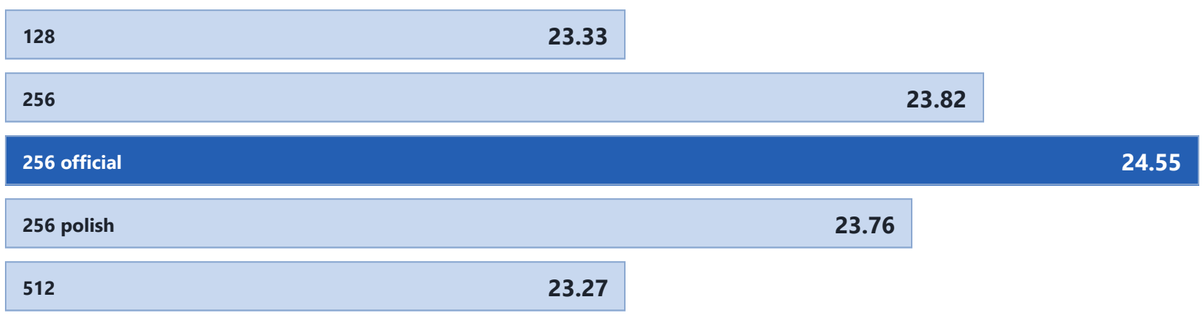
4. SIDL Paper Baseline Comparison

AirNet and DiffUIR are SIDL paper test-set values. D3Net is our official leaderboard submission using the epoch-73 checkpoint.

METHOD	LEVEL	CLEAN	DUST	FINGER	WATER	SCRATCH	MIXED	AVG.
AirNet	Easy	27.10/0.9329	23.40/0.8669	25.10/0.8708	24.85/0.8658	25.84/0.8917	25.30/0.8422	25.26/0.8784
	Medium	26.13/0.8899	19.69/0.7577	22.53/0.8191	20.36/0.7656	21.24/0.8051	16.79/0.7542	21.12/0.7986
	Hard	23.79/0.8527	16.18/0.7350	15.83/0.6432	16.64/0.7449	18.13/0.7975	14.48/0.7047	17.50/0.7463
DiffUIR	Easy	34.51/0.9785	26.96/0.9276	28.82/0.9331	27.42/0.9220	30.14/0.9375	28.22/0.9021	29.34/0.9335
	Medium	33.25/0.9375	22.11/0.8325	26.16/0.8856	24.65/0.8647	27.07/0.9105	20.32/0.8368	25.36/0.8786
	Hard	29.47/0.9366	20.38/0.8602	18.93/0.7763	19.91/0.8500	21.27/0.8599	18.97/0.8409	21.71/0.8533
D3Net ours	Easy	33.20/0.9603	25.10/0.8926	26.95/0.8994	26.45/0.8915	28.84/0.9123	27.56/0.8745	28.02/0.9051
	Medium	30.57/0.9132	22.53/0.7980	25.50/0.8554	23.73/0.8207	25.69/0.8620	20.18/0.7950	24.70/0.8407
	Hard	27.63/0.9033	20.21/0.8029	18.09/0.6940	19.47/0.7950	21.32/0.8341	18.82/0.8089	20.92/0.8064

5. Our Ablation

RUN	INIT.	CROP	EPOCH	PSNR	SSIM
128 baseline	scratch	128	34	23.33	0.8357
256 main	scratch	256	34	23.82	0.8398
256 official	latest	256	73	**24.55**	**0.8507**
256 low-LR	256 best	256	3	23.76	0.8409
512 refinement	256 best	512	2	23.27	0.8336



6. Failure Analysis

TYPE	AVG. PSNR	AVG. SSIM
Clean	30.47	0.9256
Dust	22.61	0.8312
Finger	23.51	0.8163
Water	23.22	0.8357
Scratch	25.28	0.8695
Mixed	22.19	0.8261

Hard Finger: **18.09 / 0.6940**
Hard Mixed: **18.82 / 0.8089**

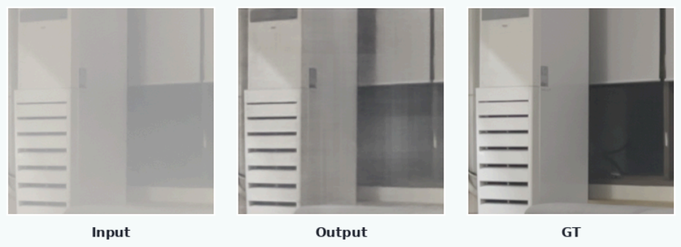
The official test result is strongest on Clean and easiest splits. Hard Fingerprint, Mixed, and Water remain the main failure regions.

7. Qualitative Results

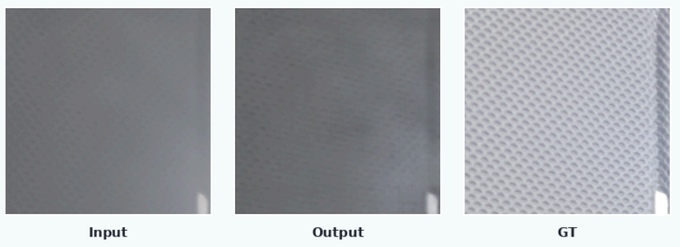
Each panel shows input, D3Net output, and GT from the report assets.

1. finger / hard / Case051_F_001.png PSNR 19.28 dB SSIM 0.9316

2. dust / medium / Case251_D_018.png PSNR 19.25 dB SSIM 0.7943



3. scratch / hard / Case073_S_001.png PSNR 10.47 dB SSIM 0.6080



4. water / hard / Case051_W_001.png PSNR 17.70 dB SSIM 0.8999

5. mixed / medium / Case170_DS_028.png PSNR 21.41 dB SSIM 0.9282

8. Takeaways

Official result
Epoch 73 reaches 24.55 dB / 0.8507 on the leaderboard test set.

Model scale
The submission reports 771 GMACs and 43.8M parameters.

Future direction
Loss-metric mismatch and difficulty-aware training remain open issues.